

WILBERT BIAO-HUI TAN

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Research Interests: Computational decoding and modeling of perception, cognition, and behavior from neural and behavioral data, toward understanding brain function and engineering systems that support and restore it across clinical and assistive settings.

EDUCATION

National University of Singapore (NUS)

2016 – 2021

B.Eng. (Honors), Electrical Engineering

- University Scholars Programme (USP): selective honors program emphasizing critical inquiry and cross-disciplinary depth; admits under 1% of each cohort (~200 of 38,000 students)
- Honors Thesis: *Spin-wave approaches towards a spiking neural network implementation*; supervised by Prof. Kelvin Xuanyao Fong;
- Coursework: Signal Processing, Biology of Perception, Biophysics, Computer Vision, Embedded Systems

Korea University, South Korea

2018

Winter Exchange Program

- Coursework: Abnormal Psychology, Business Ethics

Fudan University, China

2015

Chinese Language Program (1 Year)

- Mandarin Chinese

RESEARCH EXPERIENCE

Independent Research

2025 – Present

Neural decoding of face perception, human ECoG (Neuromatch Computational Neuroscience)

- Extracted broadband gamma from face-selective posterior fusiform recordings and identified high-SNR channels using PCA and per- and aggregate-channel analysis (Python, MNE)
- Related neural signal-to-noise ratio to behavioral face/house discrimination (nonparametric sensitivity, A')
- Found both neural and behavioral performance dropped beyond roughly 40% stimulus noise

AI/ML Foundations

- **Gaze tracking:** built a multi-stage pipeline (Python, MediaPipe, YOLO) covering calibration, real-time inference, and accuracy evaluation across variable head pose and lighting, translating gaze into on-screen control
- **Mechanistic interpretability:** studying foundational and recent AI/ML literature with a focus on circuit-level model analysis, low dimensional reduction methods, and mathematical derivation
- **RAG pipeline:** developed an end-to-end Retrieval-Augmented Generation system for deep research report generation, including a web crawler for real-time market data, news, and sentiment with deterministic quoting

Ethnographic Field Researcher, Bai Xian Institute

2022

- Interviewed Chinese “comfort women”, survivors of wartime sexual violence, family members, and researchers. Built and sustained trust across language and cultural barriers on deeply sensitive personal history

Neuromorphic Computing Researcher, SEEDER Group NUS

2020 – 2021

Supervised by Prof. Kelvin Xuanyao Fong

- Designed a spin-wave spiking neural network with spike-timing-dependent plasticity, building Mumax3 (Go) micromagnetic simulations of spin-wave propagation and plasticity run on high-performance clusters

Undergraduate Researcher, Engineering Biology Lab NUS

2018 – 2019

Supervised by Prof. Poh Chueh Loo

- Investigated bioelectric signal interfacing between bacterial systems and computational platforms for glucose monitoring
- Classified physiological time-series by encoding signals as images (Gramian Angular Field, Markov Transition Field) and training a ResNet

Computational Modelling Lead, NUS iGEM Biosafety Containment

2017

- Led the inaugural NUS iGEM team's ODE-based modeling (MATLAB) of a biological kill-switch to guide wet-lab design; awarded Gold Medal among 310 teams worldwide

APPLIED & INDUSTRY EXPERIENCE

Customer Success Manager (AI), US Financial Services, IBM 2024

- **AI strategy advisory:** served as technical advisor to Fortune 500 financial services clients on AI strategy, data governance, and LLM implementation
- **Cross-functional delivery:** Partnered with data science, sales, and solution architecture teams to scope, design and position AI solutions against competing vendors, driving client adoption across data governance and LLM initiatives.

Data Scientist, IBM Expert Labs 2022 – 2024

- **Enterprise AI/ML:** operated across research, engineering, and client advisory functions to design and deploy first-of-a-kind solutions across the sustainability industry spanning anomaly detection, deep learning, survival modeling, reinforcement learning, time-series forecasting, mechanistic modeling, and computer vision
- Led development of transformer-health analytics model "recipes," reusable model templates integrated into IBM Maximo and deployed to over 1,000 enterprise customers as the primary code contributor.

Lead Firmware Engineer, GogoTech 2021 – 2022

- **Smart wheelchair firmware:** led embedded control system development for an affordable device funded by the Melinda Gates Foundation and TechStars
- Deployed lightweight computer vision models for real-time on-device navigation and obstacle avoidance

Electrical Controls Researcher, Rolls-Royce R&D Systems Integration 2019

Supervised by: Prof. Sanjib Kumar Panda & Dr. Souvik Dasgupta

- **Predictive microgrid control:** Developed predictive droop-control algorithms and reliable network sharing for microgrid power management using a network of programmable logic controller (C/C++, OPCUA, CAN, TCP/IP)

COMMUNITY & PARTICIPANT ENGAGEMENT

Pro-Bono Consultant, Conjoint Consulting, NUS 2020

- Developed volunteer-engagement strategies for a youth mental-wellness social enterprise in Singapore

Educational Support Volunteer, Little Beans, Shanghai 2015

- Taught weekly English to children from migrant families and facilitated learning and play-based activities for students with special needs

Educational Support Volunteer, Mudgeeraba Special School, Australia 2014

- Facilitated weekly learning and play activities for non-verbal students; organized a fundraiser for a disability-accessible bus

SELECTED PROJECTS

Electrical Engineer Lead, Singapore Medical Grand Challenge 2019

- **Wearable gait device:** designed and deployed a fall-risk assessment system (ESP32, Bluetooth, C, Altium) across multiple Singapore eldercare facilities, enabling remote clinical assessment for physical therapists. Awarded Second Prize and Marketing Prize, totalling SGD \$20,000

Team Lead, CNES-ESA ActinSpace Challenge 2018

- **VLM micro-thruster:** developed a 3D-printed water-vapor micro-thruster prototype for low-cost, environmentally friendly cubesat propulsion. Awarded Second Prize in the CNES-ESA global competition (Toulouse), the Airbus Defence & Innovation Prize, and the Singapore Space & Technology Regional Prize

Co-Founder, NomNomSnap 2016

- **Computer vision API:** built an image classification system integrated with nutritional databases for diet tracking; awarded First Prize at the Australian Health Startup Competition

Team Member, SpaceX Hyperloop Pod Competition (KeioAlpha) 2015

- Selected as one of 27 finalist teams from 120+ international applicants to build and test a Hyperloop prototype at SpaceX headquarters, Hawthorne, CA

AWARDS & HONORS

- Bai Xian Institute Project Award (\$4,000 USD): Wartime Sexual Violence and the Everydayness "Comfort Women" and Others in South Korea and China (2022)

- Singapore Medical Grand Challenge: Second Prize (SGD \$15,000) and Marketing Prize (SGD \$5,000) — Wearable Gait Analysis and Fall Prediction (2019)
- CNES-ESA ActinSpace: Global Second Prize, Airbus Defence & Innovation Prize, and Singapore Regional First Prize, for a 3D-printed water-vapor microthruster (2018)
- International Genetically Engineered Machines (iGEM): Gold Medal — Making Engineering of Customized Kill Switches Easier (2017)
- Australian Health Startup Competition: First Prize — NomNomSnap Computer Vision System for Dietary Assessment (2016)
- SpaceX Hyperloop Competition: Top 27 Finalist (out of 120+ teams worldwide) — KeioAlpha Hyperloop Team (2015)

PUBLICATIONS & PRESENTATIONS

Publications

- Tan, W. (2021). Spin-wave approaches towards a spiking neural network implementation.
B.Eng. Honors Thesis, National University of Singapore.

Presentations

IBM Maximo Models for Electrical Transformers	IBM TechXchange, 2024
Reinforcement Learning for Complex Job Shop Scheduling	IBM TechXchange, 2023
Failure Prediction in Electrical Transformer Asset and Time Series Data	IBM TechXchange, 2023
Wearable for Gait Analysis & Fall Prediction	Yong Loo Lin School of Medicine, 2019
InsertSpace: Equipping Nanosatellites with 3D-Printed VLM Thrusters	Toulouse Space Center, France, 2018
Making the Engineering of Customisable Kill-Switches Easier	iGEM Jamboree, Boston, 2017
KeioAlpha Hyperloop Poster Presentation	SpaceX Hyperloop Competition, 2016

SKILLS

Programming: Python, Linux, Bash, Go, PyTorch, scikit-learn, MNE, HuggingFace, FastAPI, Pydantic, Docker, Kubernetes, Git

Other: C/C++, Java, MATLAB, Verilog/VHDL, Assembly, OpenAI Gym

Languages: English (native), Mandarin (Fluent), Korean and Japanese (Beginner)